I. Choice (3 points X 15)

1. There are the two major components of the visual pathway from the retina to V1: P and M. What information is **NOT** carried through the M pathway, with large receptive fields?
(A) Motion
(B) High acuity
(C) Contrast
(D) Luminance

2. What is quantal release of neurotransmitter?
(A) Release of neurotransmitter in discrete units
(B) Release of one neurotransmitter molecule at a time
(C) Release of transmitter onto extrasynaptic receptors

3. Which protein is the Ca2+ sensor in synaptic transmission?
(A) Synapsin
(B) Munc18
(C) Synaptotagmin
(D) Synaptobrevin

4. If you stimulate a glutamatergic presynaptic neuron and record the response in the postsynaptic neuron with only NMDA receptors, what response will you get?
(A) Depolarizing
(B) Hyperpolarizing
(B) No response

5. Why does the NMDAR provide a good molecular mechanism for coincidence detection? Select the best answer.
(A) It is the receptor for glutamate, the major excitatory transmitter in the brain.
(B) It requires depolarization to remove the Mg2+ block.
(C) It allows Ca2+ into the postsynaptic neuron.
(D) The NMDAR channel can remain open for long periods of time.
(E) It is expressed in the hippocampus.

6. What is the Output nucleus of the basal ganglia?
(A) Striatum
(B) VTA
(C) Substantia nigra pars reticulata (SNr)
(D) Subthalamic nucleus (STN)
(E) Globus pallidus

7. Parkinson's disease is directly related to which neurotransmitter?
(A) Glutamate
(B) GABA
(C) Epinephrine
(D) Dopamine

8. Language is processed in which brain area?
(A) Hypothalamus
(B) Thalamus
(C) Broca's area
(D) Brain stem

9. What is the specialty of the node of Ranvier?
(A) There is a high concentration of Na+ channels.
(B) There is a high concentration of K+ channels.
(C) This is the point at which most synaptic contacts are made.
(D) It is located at the beginning of the axon.

10. Which ion is directly related to vesicle release?
(A) Na+
(B) K+
(C) Ca2+
(D) Cl-

11. To which areas of the brain do olfactory sensory neurons directly project?
A. LGN
B. Olfactory cortex
C. Olfactory thalamus
D. Olfactory bulb

12. What would happen if you lesioned the left LGN?
A. You would not be able to see the right visual field.
B. You would not be able to see the left visual field.
C. You would not receive any information from your right eye (functionally blind in the right eye).
D. You would not receive any information from your left eye (functionally blind in the left eye).

13. What are the two most common neurotransmitters in the autonomic nervous system?
A. GABA and acetylcholine
B. Glycine and acetylcholine
C. Norepinephrine and acetylcholine
D. GABA and glutamate

14. AMPA receptor is what kind of protein?
A. Voltage-dependent ion channel
B. ligand-gated ion channel
C. Na+ pump
D. G-protein-coupled receptor

15. What do you predict would happen if the activity of hypocretin neurons in the hypothalamus were optogenetically activated?
A. The animal would spend more time asleep.
B. The animal would have an increased probability of making a transition from sleep to wake.
C. The animal would not be able to wake at all.
D. The animal would spend more time in NREM sleep.

II. Short Answer (5 points X 5)

1. Which is spike-timing-dependent plasticity?

2. How does ChR2 activate neurons?

3. Dopamine has opposing effects on the direct and indirect pathways in the basal ganglia. How is this possible?

4. What is action potential?

5. Summarize the key principles of the projection pattern of the olfactory system.

III. Essay questions (15 points X 2)

1. A student found that the level of a GPCR, SLST2022, is dramatically decreased in the prefrontal cortex of the brain of depressive mice. He later identified the ligand of SLST2022 as a neuropeptide Lucky2022.

Design experiments to test the role of SLST2022 and Lucky2022.

2. Describe the response properties of neurons in the motor cortex. Based on such properties, how could you design a brain-machine interface to perform motor functions?